# Washing program listing

ROM addresses are hexadecimal instruction-word addresses. Each row maps the assembled program to its exact 16-bit machine word. Unlisted ROM words 6E through FF are zero NOPs.

| Address | Hex word | Assembly |
|---|---|---|
| 00 | 10F0 | LDI R0, 0xF0 |
| 01 | 1609 | LDI R3, 0x09 |
| 02 | 6200 | LD R1, [R0] |
| 03 | 2440 | MOV R2, R1 |
| 04 | 52C0 | AND R1, R3 |
| 05 | 42C0 | SUB R1, R3 |
| 06 | A202 | BNEZ R1, 0x02 |
| 07 | 1602 | LDI R3, 0x02 |
| 08 | 54C0 | AND R2, R3 |
| 09 | A40E | BNEZ R2, 0x0E |
| 0A | 1864 | LDI R4, 0x64 |
| 0B | 1E05 | LDI R7, 0x05 |
| 0C | 160C | LDI R3, 0x0C |
| 0D | B011 | JMP 0x11 |
| 0E | 1896 | LDI R4, 0x96 |
| 0F | 1E08 | LDI R7, 0x08 |
| 10 | 1610 | LDI R3, 0x10 |
| 11 | 1AF9 | LDI R5, 0xF9 |
| 12 | 1C01 | LDI R6, 0x01 |
| 13 | 7D40 | ST R6, [R5] |
| 14 | 1AF5 | LDI R5, 0xF5 |
| 15 | 1C00 | LDI R6, 0x00 |
| 16 | 7D40 | ST R6, [R5] |
| 17 | 1AF4 | LDI R5, 0xF4 |
| 18 | 1C01 | LDI R6, 0x01 |
| 19 | 7D40 | ST R6, [R5] |
| 1A | 1AF1 | LDI R5, 0xF1 |
| 1B | 6340 | LD R1, [R5] |
| 1C | D300 | CMPGEU R1, R4 |
| 1D | A21F | BNEZ R1, 0x1F |
| 1E | B01A | JMP 0x1A |
| 1F | 1800 | LDI R4, 0x00 |

00–06 wait for START and door; 07–10 choose parameters; 11–19 lock and start fill. Targets are absolute word addresses.

## Program listing continued

| Address | Hex word | Assembly |
|---|---|---|
| 20 | 1AF4 | LDI R5, 0xF4 |
| 21 | 7940 | ST R4, [R5] |
| 22 | 1AFD | LDI R5, 0xFD |
| 23 | 7F40 | ST R7, [R5] |
| 24 | 1AF7 | LDI R5, 0xF7 |
| 25 | 1C00 | LDI R6, 0x00 |
| 26 | 7D40 | ST R6, [R5] |
| 27 | 1AFE | LDI R5, 0xFE |
| 28 | 1203 | LDI R1, 0x03 |
| 29 | 14F6 | LDI R2, 0xF6 |
| 2A | 1C01 | LDI R6, 0x01 |
| 2B | 7340 | ST R1, [R5] |
| 2C | 7C80 | ST R6, [R2] |
| 2D | 1AFF | LDI R5, 0xFF |
| 2E | 6340 | LD R1, [R5] |
| 2F | A231 | BNEZ R1, 0x31 |
| 30 | B02E | JMP 0x2E |
| 31 | 7880 | ST R4, [R2] |
| 32 | 1AFE | LDI R5, 0xFE |
| 33 | 7940 | ST R4, [R5] |
| 34 | 1AFD | LDI R5, 0xFD |
| 35 | 7F40 | ST R7, [R5] |
| 36 | 1AF7 | LDI R5, 0xF7 |
| 37 | 1C01 | LDI R6, 0x01 |
| 38 | 7D40 | ST R6, [R5] |
| 39 | 1AFE | LDI R5, 0xFE |
| 3A | 1203 | LDI R1, 0x03 |
| 3B | 14F6 | LDI R2, 0xF6 |
| 3C | 1C01 | LDI R6, 0x01 |
| 3D | 7340 | ST R1, [R5] |
| 3E | 7C80 | ST R6, [R2] |
| 3F | 1AFF | LDI R5, 0xFF |

The timer is cleared and rearmed for each wash direction. R4 holds zero after filling, and R3 retains the spin limit.

## Program listing continued

| Address | Hex word | Assembly |
|---|---|---|
| 40 | 6340 | LD R1, [R5] |
| 41 | A243 | BNEZ R1, 0x43 |
| 42 | B040 | JMP 0x40 |
| 43 | 7880 | ST R4, [R2] |
| 44 | 1AFE | LDI R5, 0xFE |
| 45 | 7940 | ST R4, [R5] |
| 46 | 1AF8 | LDI R5, 0xF8 |
| 47 | 1C01 | LDI R6, 0x01 |
| 48 | 7D40 | ST R6, [R5] |
| 49 | 1AF1 | LDI R5, 0xF1 |
| 4A | 6340 | LD R1, [R5] |
| 4B | 140A | LDI R2, 0x0A |
| 4C | D440 | CMPGEU R2, R1 |
| 4D | A44F | BNEZ R2, 0x4F |
| 4E | B049 | JMP 0x49 |
| 4F | 1AF8 | LDI R5, 0xF8 |
| 50 | 7940 | ST R4, [R5] |
| 51 | 1AFD | LDI R5, 0xFD |
| 52 | 7740 | ST R3, [R5] |
| 53 | 1AF7 | LDI R5, 0xF7 |
| 54 | 1C00 | LDI R6, 0x00 |
| 55 | 7D40 | ST R6, [R5] |
| 56 | 1AFE | LDI R5, 0xFE |
| 57 | 1203 | LDI R1, 0x03 |
| 58 | 14F6 | LDI R2, 0xF6 |
| 59 | 1C01 | LDI R6, 0x01 |
| 5A | 7340 | ST R1, [R5] |
| 5B | 7C80 | ST R6, [R2] |
| 5C | 1AFF | LDI R5, 0xFF |
| 5D | 6340 | LD R1, [R5] |
| 5E | A260 | BNEZ R1, 0x60 |
| 5F | B05D | JMP 0x5D |

## Program listing continued

| Address | Hex word | Assembly |
|---|---|---|
| 60 | 7880 | ST R4, [R2] |
| 61 | 1AFE | LDI R5, 0xFE |
| 62 | 7940 | ST R4, [R5] |
| 63 | 1AF9 | LDI R5, 0xF9 |
| 64 | 7940 | ST R4, [R5] |
| 65 | 1AFB | LDI R5, 0xFB |
| 66 | 1C01 | LDI R6, 0x01 |
| 67 | 7D40 | ST R6, [R5] |
| 68 | 1AFA | LDI R5, 0xFA |
| 69 | 7D40 | ST R6, [R5] |
| 6A | 1AF7 | LDI R5, 0xF7 |
| 6B | 1C00 | LDI R6, 0 |
| 6C | 7D40 | ST R6, [R5] |
| 6D | C000 | HALT |

After draining, the timer is rearmed for spin. Final stores stop the motor, disable the timer, unlock, indicate completion and clear direction before HALT at 6D.